## 10.10 THE ELEMENTS OF AN ELECTRONIC COMMUNICATION SYSTEM

Electronics plays a major role in communication. Electronic communication is nothing but the transmission of sound, text, pictures, or data through a medium. Long distance transmission uses free space as a medium.

The elements of the basic communication system are explained with the block diagram shown in Figure 10.52.

### i) Information (Baseband or input signal)

Information can be in the form of speech, music, pictures or computer data. This information is given as input to the input transducer.

### ii) Input transducer

In communication system, the transducer converts the information (in the form of sound, music, pictures or computer data) into corresponding electrical signals. **The electrical equivalent of the original information is called the baseband signal.**

The best example for the transducer is the microphone that converts sound energy into electrical energy.

### iii) Transmitter

It feeds the electrical signal from the transducer to the communication channel. It consists of circuits such as amplifier, oscillator, modulator and power amplifier. The transmitter is located at the broadcasting station.

**Amplifier**: The transducer output is very weak and is amplified by the amplifier.

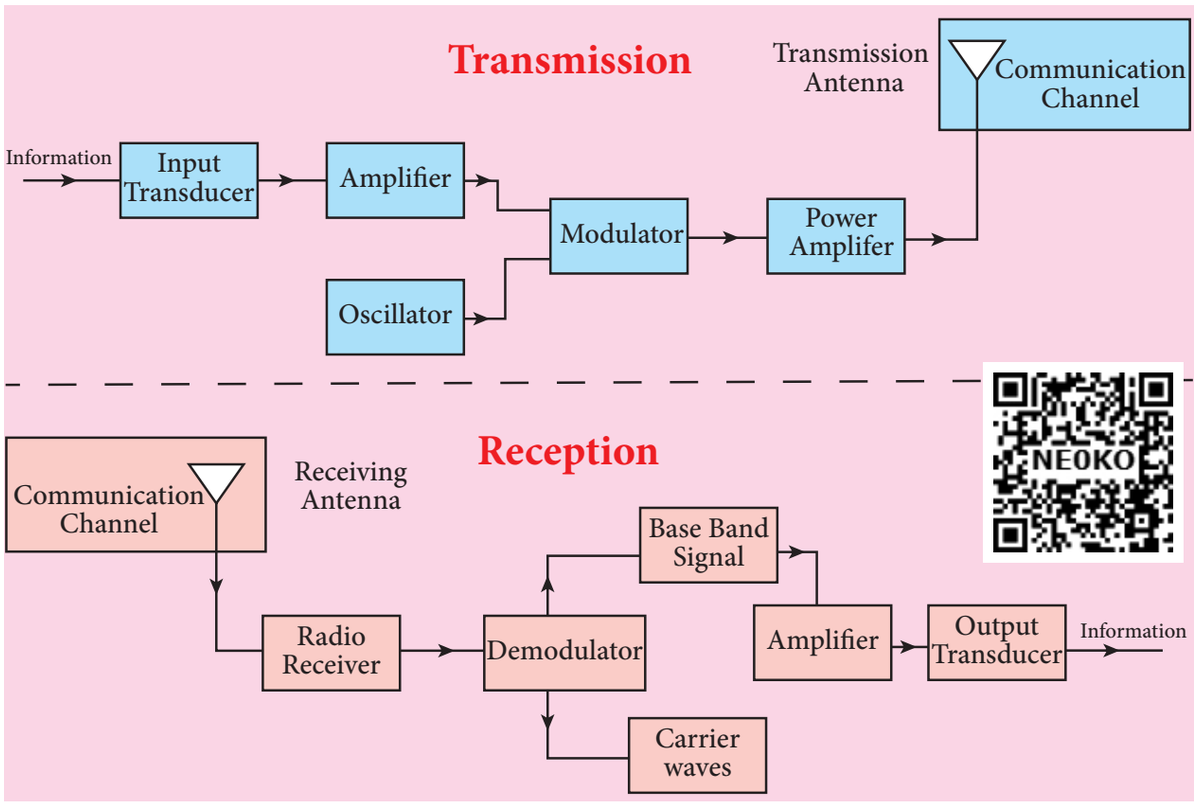



**Figure 10.52** Block diagram of transmission and reception of voice signals



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